Ch. 8 Relationship between exchange rates, interest rates and inflation

IRP = interest rate parity

$$\frac{F}{S} - 1 = p = i_h - i_f$$

Fisher

nominal i = I + r

I = inflation

r = real interest rate

Change in purchasing power

Purchasing Power Parity

Given US and Great Britain, if US experiences higher inflation, US consumers will purchase more abroad, thus increasing demand for British Pound and increasing supply

↓ USD → GBP↑, USD↓

→ (Absolute) Purchasing Power Parity does not account for tariffs and transaction costs.

→ Relative Purchasing Power Parity

→ rate of change in prices between two countries are determined by inflation differences

PPP

$$\%\Delta EX = \overline{Infl._h} - Infl._f$$

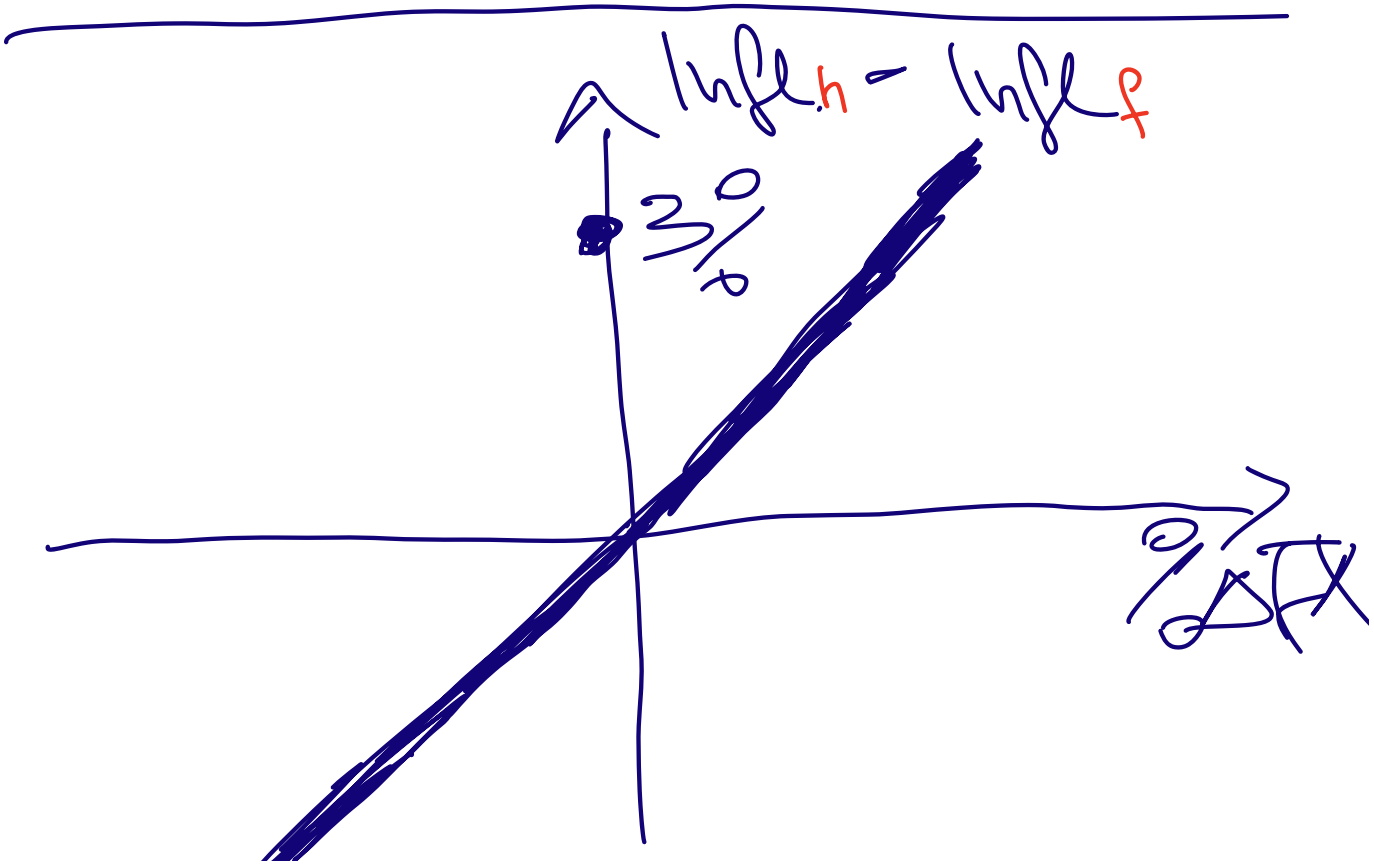
if British pound was trading at $1.50 last month and today its exchange rate is $1.60/£, verify whether relative purchasing power

Parity holds if US inflation rate is 4% and British inflation rate is 6%.

$$\%\Delta FX = I_{infl.h} - I_f$$

$$\frac{\$1.60}{\$1.50} - 1 = 4\% - 6\%$$

$$6.67\% \neq -2\%$$



$Inf_h - Inf_f = 3\%$

$i = \overline{Inf.} + \text{real rate}$

International Fisher Effect

→ real growth is fairly stable at a country level

→ we can infer inflation from nominal interest rates and real rates

Use International Fisher Effect (IFE) to infer current inflation to estimate

purchasing power parity